Dental Vision Benchmark: A Clinician-Reviewed, Contamination-Aware Evaluation of Vision Language Models on Non-Radiographic Dental Images

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Abstract

Clinicians already show vision language models a diagram, an illustration, or a clinical photograph and ask "what is this?", but no public benchmark measures which model reads a dental image correctly when radiographs are excluded. We present a small, clinician-authored, contamination-aware benchmark of *non-radiographic* dental image understanding: 90 openly licensed images (52 diagrams and 38 clinical photographs) across roughly sixteen categories, each paired with explicit *must identify* and *must avoid* rubric criteria and a reference caption. Six contemporary vision models were evaluated through OpenRouter on June 15, 2026, with the caption stripped and no tool or web access (a leakage guard), and each free-text description was graded by two independent judges from different vendors that each see the image and the rubric. Gemini 3.1 Pro led at 72% (95% Wilson interval [62, 80]), followed by Claude Opus 4.8 at 63% [53, 73] and Qwen3.7 Plus at 61% [51, 71]; these intervals overlap and the ordering is not statistically resolved. The robust finding is a task split rather than a ranking: every model reads diagrams and labeled images far better than unlabeled clinical photographs, where the weaker models fall to 20–29%. The two judges agreed on 82% of descriptions (Cohen's $\kappa = 0.65$), so absolute pass rates are judge-dependent. Ground truth was reviewed by the periodontist author and independently confirmed by a second dentist (90/90 items judged clinically fair). A transformed-control item showed no evidence that exact-image recognition drives scores. The images, rubrics, per-row results, judge verdicts, and analysis code are released for reproducible, specialty-specific evaluation.

1 Introduction

Vision language models are now routinely shown medical images and asked to interpret them. General medical multimodal benchmarks have grown quickly, but they favor breadth and a closed question-answering format: OmniMedVQA spans 12 imaging modalities and more than 118,000 images, and reports that most models only slightly exceed random guessing [1]; GMAI-MMBench is built from 284 datasets across 38 modalities, where even a leading proprietary model reached only 53.96% [2]. These suites measure option elimination at scale, not whether a model produces a clinically faithful free-text reading of a single image.

Dentistry is underrepresented in this literature, and the dental work that does exist is dominated by *radiographs*. OralMLLM-Bench evaluates models across periapical, panoramic, and lateral cephalometric radiographs and documents a persistent gap between models and clinicians [3]. Large oro-dental datasets pair intraoral images and radiographs with clinical records [4], and text-centric efforts such as DentalBench measure factual dentistry knowledge rather than image reading [5]. The closest analog to the present work evaluated six model endpoints on oral and maxillofacial anatomy using landmark identification on atlas plates, where the best endpoint reached 53.1% and the authors concluded that no model is reliable enough to serve as a stand-alone answer key [6].

Non-radiographic dental image reading, the diagrams, illustrations, and clinical photographs a clinician or student encounters daily, is its own competency, and it is not covered by the benchmarks above. We therefore contribute a deliberately small, fully public, clinician-authored benchmark with three design commitments that the general medical and dental benchmarks do not jointly provide: open-ended free-text grading against an explicit clinician rubric, a leakage guard that removes the caption and any web access at input, and a transformed-control check for exact-image recognition. We treat the model judge as an empirical quantity, reporting inter-judge agreement rather than assuming it, in line with work on LLM-as-judge reliability [7, 8] and self-preference bias [9, 10].

2 Methods

2.1 Benchmark dataset

The benchmark contains 90 dental images: 52 diagrams or illustrations and 38 clinical photographs, across roughly sixteen categories (tooth and periodontal anatomy, periodontology, caries, restorative dentistry, endodontics, implants, orthodontics and occlusion, oral pathology and mucosal lesions, developmental anomalies, tooth wear, trauma, and temporomandibular or salivary structures). Radiographic interpretation is deliberately out of scope; it is a separate, harder competency and is treated as a later tier. Forty-one images carry readable text labels and 49 do not. All images are openly licensed (Wikimedia Commons or open-access sources); per-image attribution is released with the dataset.

Each image is paired with a clinician-authored rubric rather than its original web caption: a set of *must identify* points that a correct reading must cover, a set of *must avoid* errors, and a reference caption. A description is scored correct only if it covers all required points and commits no prohibited error. Disease severity, when applicable, is recorded but reported as a soft metric and does not gate the pass/fail score.

2.2 Models and execution

We evaluated six vision-capable models, the strongest of each model family available on OpenRouter on the run date (Table 1). Two strong open-weight releases of the period, GLM-5.2 and Rio 3.5, are text-only and therefore cannot enter a vision benchmark. Each image was shown to each model once at temperature 0, with a 1,500-token description budget. OpenRouter routed each request to an upstream provider: Claude Opus 4.8, GPT-5.5, Qwen3.7 Plus, and GLM-4.6V resolved to a single provider each (Anthropic, OpenAI, Alibaba, and Novita respectively), Gemini 3.1 Pro was served almost entirely by Google, and Llama 4 Maverick was distributed across four providers (Parasail, Google, DeepInfra, and Novita). The per-row provider is recorded in the released results and the run manifest. The run produced 90 images $\times$ 6 models $=$ 540 description rows.

Table 1: Model roster. Identifiers are the OpenRouter identifiers recorded by the harness on the run date.

Label	OpenRouter model identifier	Vendor
Gemini 3.1 Pro	`google/gemini-3.1-pro-preview`	Google
Claude Opus 4.8	`anthropic/claude-opus-4.8`	Anthropic
Qwen3.7 Plus	`qwen/qwen3.7-plus`	Alibaba
GPT-5.5	`openai/gpt-5.5`	OpenAI
GLM-4.6V	`z-ai/glm-4.6v`	Zhipu AI
Llama 4 Maverick	`meta-llama/llama-4-maverick`	Meta

2.3 Leakage guard

Each model received only the image, encoded inline as a base64 data URI, with the caption removed and with no tools or web access. Because the model cannot retrieve the original caption, a high score can only come from reading the picture rather than recalling associated text. The caption is withheld from the contestant but is given to the judges as ground truth, so this guard limits input-side recall; it does not by itself certify that a passing description was read from the pixels. The full pipeline is shown in Figure 1.

2.4 Dual-judge rubric scoring

Two independent judges from different vendors graded each description: a primary judge (Claude Opus 4.8) and a secondary judge (GPT-5.5). Each judge was shown the real image together with the authoritative caption and the clinician rubric, and returned a structured verdict containing the number of required points covered, the total required, the number of prohibited violations, a modality-correctness flag, and a soft severity judgment. The primary judge drives the reported leaderboard; the secondary judge quantifies judge dependence. Both judges' full verdicts are stored per row.

2.5 Transformed control

A transformed public control item, a flipped-and-cropped copy of one public image, is a sanity check on exact-image recognition: a model that reads the picture should score the same on the control. This is a control, not a memorization measurement.

2.6 Outcomes and statistics

The primary outcome is rubric pass rate per model over the 90 images. For per-model uncertainty we report 95% Wilson score intervals. We report task splits (diagram versus clinical photograph, labeled versus unlabeled), modality-identification accuracy, a sensitivity analysis excluding 10 items pre-flagged as weak or ambiguous, and inter-judge agreement (raw verdict agreement and Cohen's κ). These intervals reflect a 90-image set and should not be read as population-level confidence intervals over all dental images.

2.7 Clinician validation

Ground truth was authored and reviewed by the periodontist author through a per-image confirm-by-default review (87 of 90 confirmed as authored; 3 corrected, comprising two rubric relaxations

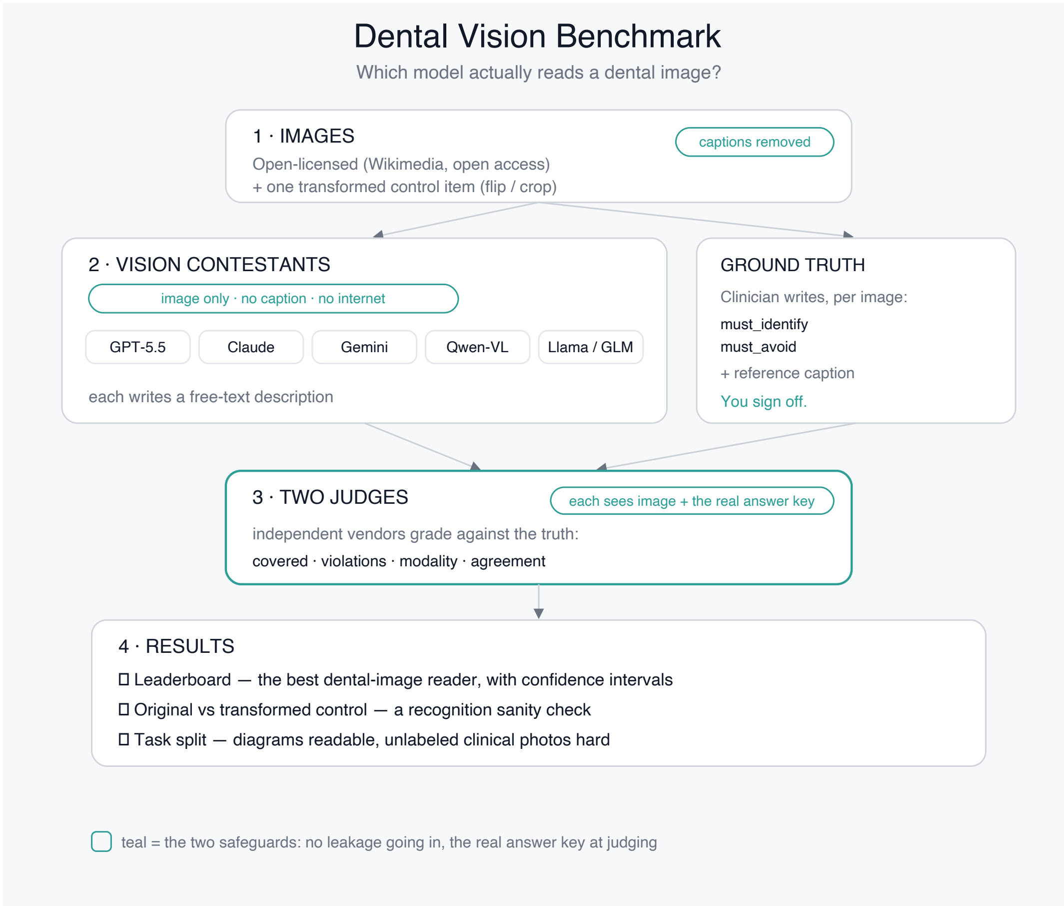

Figure 1: Benchmark pipeline. Images are shown caption-free to each vision model; two independent judges that each see the image and hold the clinician rubric grade every description; the outputs are a leaderboard with confidence intervals, a transformed-control check, and the diagram-versus-photo task split.

and one mislabeled image replaced and re-authored). A second, independent dentist (D.R.C.) then reviewed all 90 items against the released ground truth through a structured validation page and confirmed every item as clinically fair, with no rubric edits and no exclusions. The validation record is released with the repository.

3 Results

3.1 Overall pass rate

Across the 540 description rows, primary-judge pass rate was 52%. Per-model results are shown in Table 2 and Figure 2. Gemini 3.1 Pro led at 72% (65/90), followed by Claude Opus 4.8 at 63% (57/90), Qwen3.7 Plus at 61% (55/90), GPT-5.5 at 56% (50/90), GLM-4.6V at 34% (31/90), and Llama 4 Maverick at 28% (25/90).

The top three models form a cluster whose intervals overlap substantially, and this study should not be cited as evidence of a resolved ordering among them. The more informative separation is between this leading cluster and the open-weight tail. Notably, GPT-5.5 sits below the cluster

Table 2: Primary benchmark results (primary judge). Pass rate with 95% Wilson intervals over 90 images, and pass rate within each task split.

Model	Pass rate (95% CI)	Diagrams	Clinical photos	Labeled	Unlabeled
Gemini 3.1 Pro	72% [62, 80]	71%	74%	76%	69%
Claude Opus 4.8	63% [53, 73]	69%	55%	76%	53%
Qwen3.7 Plus	61% [51, 71]	67%	53%	71%	53%
GPT-5.5	56% [45, 65]	63%	45%	68%	45%
GLM-4.6V	34% [25, 45]	38%	29%	41%	29%
Llama 4 Maverick	28% [20, 38]	33%	21%	27%	29%

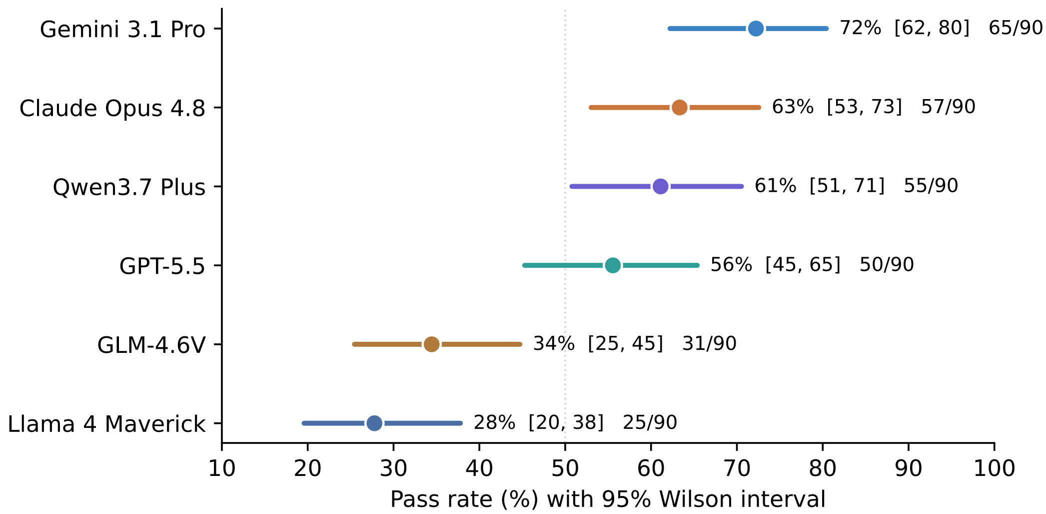

Figure 2: Pass rate by model with 95% Wilson intervals over the 90 images. The top three intervals overlap; the ordering within that cluster is not statistically resolved.

here, even though the OpenAI family is among the strongest systems on text-based clinical question answering; strength on text need not transfer to vision. A sensitivity analysis excluding the 10 pre-flagged weak or ambiguous items (n=80) shifted every model upward by 3 to 5 points and left the ranking unchanged.

3.2 The task split is the headline

Figure 3 and the right-hand columns of Table 2 show the robust result. Every model reads diagrams and labeled images better than unlabeled clinical photographs. The gap is largest for the mid and lower tier: GPT-5.5 falls from 63% on diagrams to 45% on unlabeled images, and GLM-4.6V and Llama 4 Maverick fall to 29% and below on unlabeled clinical photographs. Gemini 3.1 Pro is the exception, reading clinical photographs (74%) as well as diagrams (71%). Reading an unlabeled real mouth, rather than a textbook diagram, is the hard and still unsolved part of this task. Modality identification, by contrast, was nearly saturated: every model labeled diagram versus photograph correctly about 99% of the time, so the failures are in clinical reading, not image-type confusion.

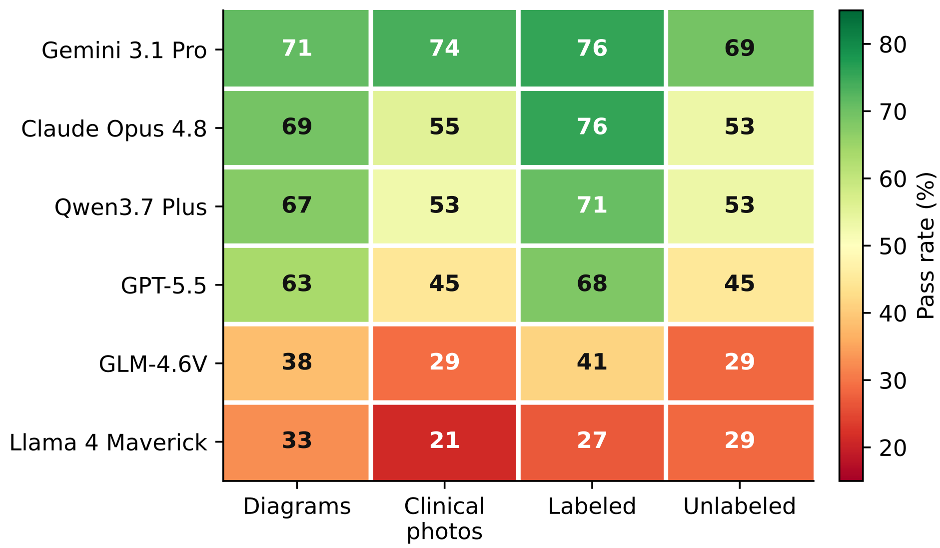

Figure 3: Pass rate by model and task split. Diagrams and labeled images are read more reliably than unlabeled clinical photographs across the field.

Table 3: Judge agreement. Pass rate assigned by the primary (Claude Opus 4.8) and secondary (GPT-5.5) judges, and their difference. Overall verdict agreement 82%, Cohen's $\kappa = 0.65$.

Model	Primary judge	Secondary judge	Δ
Gemini 3.1 Pro	72%	50%	+22
Claude Opus 4.8	63%	56%	+7
Qwen3.7 Plus	61%	47%	+14
GPT-5.5	56%	36%	+20
GLM-4.6V	34%	16%	+18
Llama 4 Maverick	28%	11%	+17

3.3 Judge agreement

The two judges returned the same pass/fail verdict on 82% of descriptions, with Cohen's $\kappa = 0.65$ (Table 3). The primary judge (Claude Opus 4.8) was more lenient than the secondary judge (GPT-5.5) for every model. Claude Opus 4.8 had the *smallest* primary-versus-secondary gap (+7 points), not the largest, which is not what a simple self-preference effect would predict. This delta is only suggestive, however: it conflates the secondary judge's overall strictness with any self-preference, and because the reported leaderboard uses the more lenient primary judge, the design cannot cleanly separate leniency from self-preference. The defensible reading is that absolute pass rates are strongly judge-dependent and should not be taken on faith.

3.4 Transformed control

The transformed control is a single item, a flipped and cropped copy of one public anatomy diagram whose original is passed by all six models, so it is a weak format-robustness sanity check rather than a memorization measurement. Under the primary judge, every model returned the same pass verdict on the control as on the original; under the secondary judge, the two weakest models (GLM-4.6V and Llama 4 Maverick) flipped from pass to fail on the transformed copy. The check is consistent with image reading rather than exact-image recognition of the public files, but a single ceiling-level item cannot establish that on its own.

3.5 Contamination and held-out evaluation

The public images are openly licensed and crawlable, so training-data exposure is a genuine threat to validity [11]. Two guards address it at the level a public dataset allows: the leakage guard at input removes the caption and any web access, and the transformed-control check found no exact-image recognition effect. Neither substitutes for a definitive contamination test, which requires a held-out set of non-public images matched in content and difficulty to the public set; constructing and running that set is the central item of future work.

4 Discussion

Contemporary vision language models can read many dental diagrams and labeled images, but this benchmark shows why specialty-specific, open-ended evaluation remains necessary. The leading systems cluster tightly with overlapping intervals, so a ranking from 90 images would be overconfident; the defensible conclusion is that a leading cluster (Gemini 3.1 Pro, Claude Opus 4.8, Qwen3.7 Plus) performed substantially better than the open-weight tail, and that the diagram-versus-photo task split, not the rank order, is the stable signal. Strength on text-based clinical question answering need not predict vision-reading rank, which cautions against transferring text leaderboards to vision tasks.

The open-ended, rubric-graded design exposes failures that closed question-answering can hide. A description can be fluent and confident yet fail because it omits a required finding or commits a prohibited error, and the unlabeled clinical-photograph column is where that happens most. For a clinician or educator deciding which model to trust with an unlabeled intraoral photograph, the right summary is not a single winner but a task-conditioned expectation: reading diagrams is largely solved, reading an unlabeled real mouth is not.

The method is intentionally small and preliminary. The benchmark uses a single trial, an LLM-judge layer whose absolute scores are judge-dependent, and ground truth from two dentists rather than a large consensus panel. The secondary-judge pass and the released judge-disagreement worklist quantify the judge problem rather than solve it; prespecified human adjudication of the disagreement set is the natural next step.

5 Limitations

First, the benchmark contains 90 images; the Wilson intervals are correspondingly wide and small differences should be treated as noise. Second, the run used a single trial at temperature 0; run-to-run stability was not measured. Third, ground truth was authored by one periodontist and confirmed by one independent dentist, which is stronger than single-author ground truth but is not a multi-expert consensus panel, and the independent reviewer reported agreement rather than adjudicating judge disagreements. Fourth, all pass/fail verdicts rely on model judges that, while in substantial agreement by the Landis–Koch scale ($\kappa = 0.65$), are themselves evaluated models, and the two judges differ enough that absolute pass rates are judge-dependent. Fifth, contamination cannot be fully excluded for openly licensed images, and a content-matched held-out evaluation is left to future work. Sixth, results reflect model versions and OpenRouter routing on June 15, 2026, which may drift. Finally, the benchmark scores text descriptions of dental images; it does not validate clinical deployment, diagnosis, or patient outcomes, and radiographic interpretation is out of scope.

6 Data and Code Availability

The images, clinician rubrics, per-row results, both judges' verdicts, the run manifest, the analysis and figure scripts, the validation record, and an interactive report are available at:

https://github.com/Tuminha/dental-vision-benchmark

The released data correspond to the pinned commit `5ccc42f`. Images are openly licensed and keep their own licenses; the code is MIT. An interactive report is at https://tuminha.github.io/dental-vision-benchmark/.

7 Ethics Statement

This study used openly licensed dental images and clinician-authored rubrics. It did not use patient records or protected health information. The results should not be interpreted as validating any model for autonomous diagnosis, treatment planning, or patient-specific clinical decision-making.

8 Author Contributions

F.T.B. designed the benchmark, curated and authored the dataset and rubrics, implemented the evaluation harness, ran the analyses, and drafted the manuscript. D.R.C. performed independent clinical validation of the ground truth and contributed clinical and scientific review.

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